



DECODE SIH 2026



India's Largest Student-Led SIH Innovation Hackathon



**SMART INDIA
HACKATHON
2026**



CITINEL

CAUGHT. CITED. CLOSED.



AEROFYTA



Presented By: Team AeroFyta
Chennai Institute of Technology



Track: Bharat Pragati

AI for Cybersecurity, FinTech
and Digital Governance



Problem Statement PS4

Autonomous Cyber SOC for AI-Powered
Threat Detection and Automated Incident Response



Every Second Counts

TEAM MEMBERS

every layer of the pipeline has a named owner.



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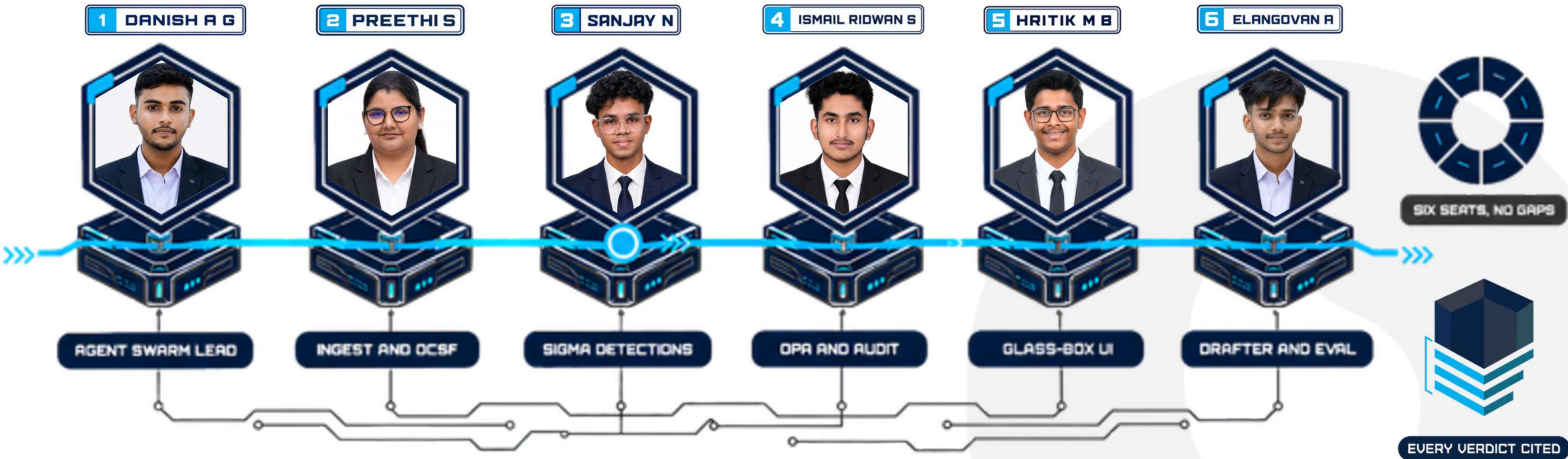


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by communityX

THE CREW IS THE PIPELINE



EVERY SEAT HAS A BACKUP

	1	2	3	4	5	6
OWNS	☒	☒	☒	☒	☒	☒
BACKS UP	☐	☐	☐	☐	☐	☐

ROUTE ANY QUESTION

EVERY QUESTION HAS AN OWNER TOO

WRONG ACTION?	SEAT 4	POISONED LOG?	SEAT 2
HALLUCINATION?	SEAT 6	WHO PAYS?	SEAT 1



WHAT EACH SEAT PASSES ON

PROBLEM STATEMENT



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in 2024, one attack took roughly 300 cooperative banks offline. the six-hour reporting clock did not pause.

WHAT IT ASKS US TO SOLVE

A security centre that runs itself: AI finds the threat, proves it with evidence, acts inside a rulebook a human can read, and drafts the mandatory **six-hour report** before the clock runs out.

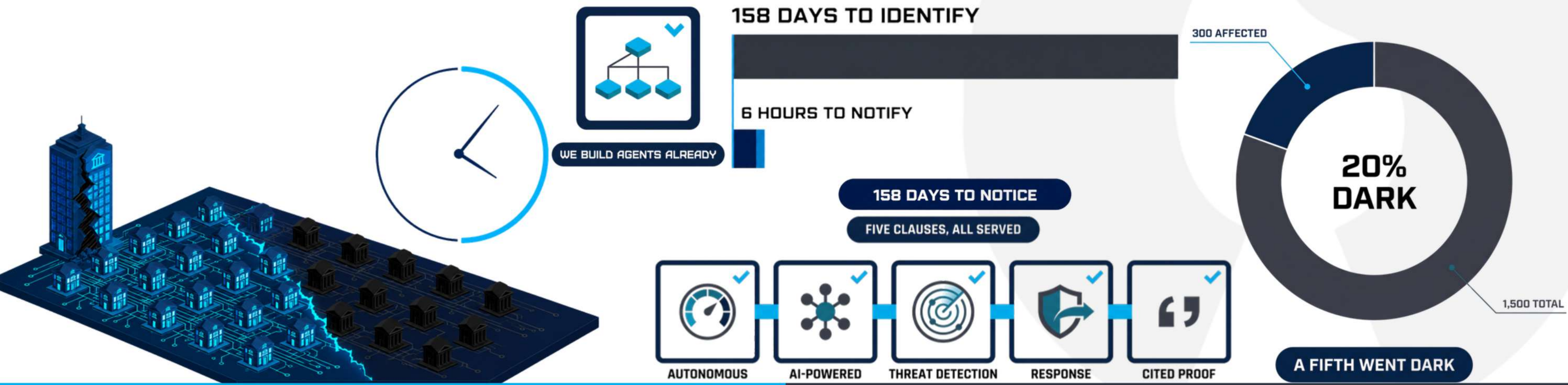
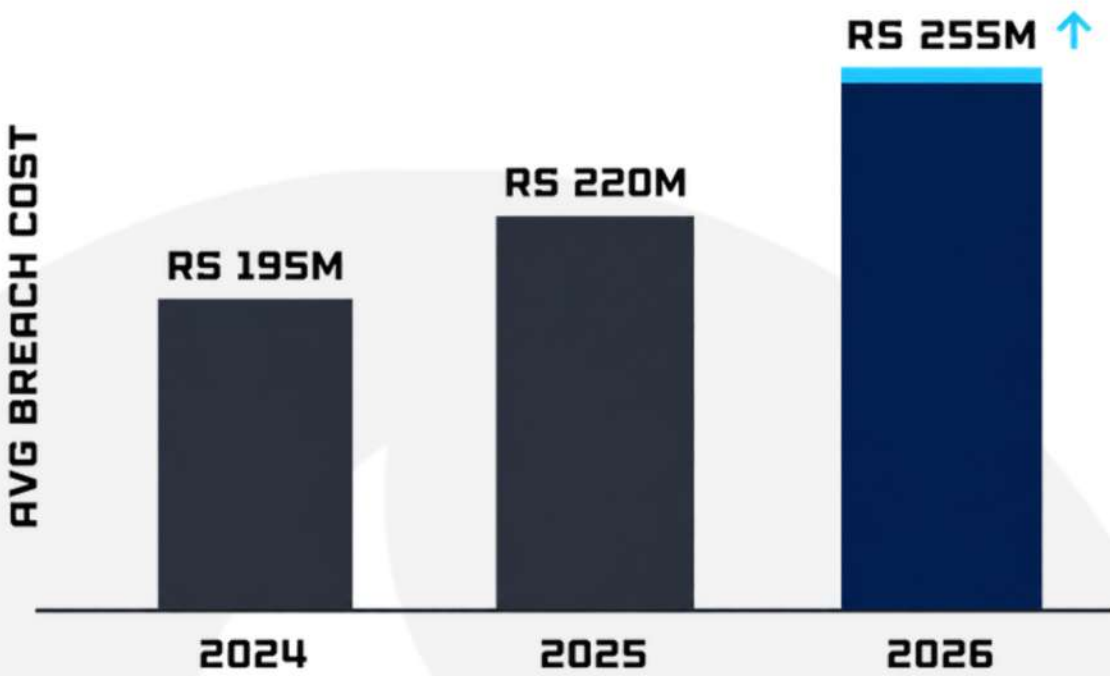
THE GAP TODAY

Detection tools stop at the alert. Compliance tools start at a **blank form**. Nothing crosses the gap so a human retypes facts the system already holds, during the six hours that matter.

WHY WE CHOSE IT

One 2024 attack took ~300 cooperative banks offline. RBI requires them to run a **24×7 SOC** that costs more than their whole **IT budget**. Our lead builds agentic systems professionally.

RS 255M, A NEW RECORD



REAL-WORLD PROBLEM ALIGNMENT

regulators mandate a 24x7 soc that costs \$1m+ a year. the banks it binds budget ₹25k to ₹1l a month.

ONE IT HEAD,
ZERO ANALYSTS



STAFFED 24X7 SOC USD 1M PLUS PER YEAR

BUYER CEILING
RS 25,000 TO RS 1 LAKH PER MONTH

RULES NOTIFIED 13 NOV 2025
CONSENT MANAGER 13 NOV 2026



RUNWAY CLOSES MAY 2027

FULLY ENFORCEABLE

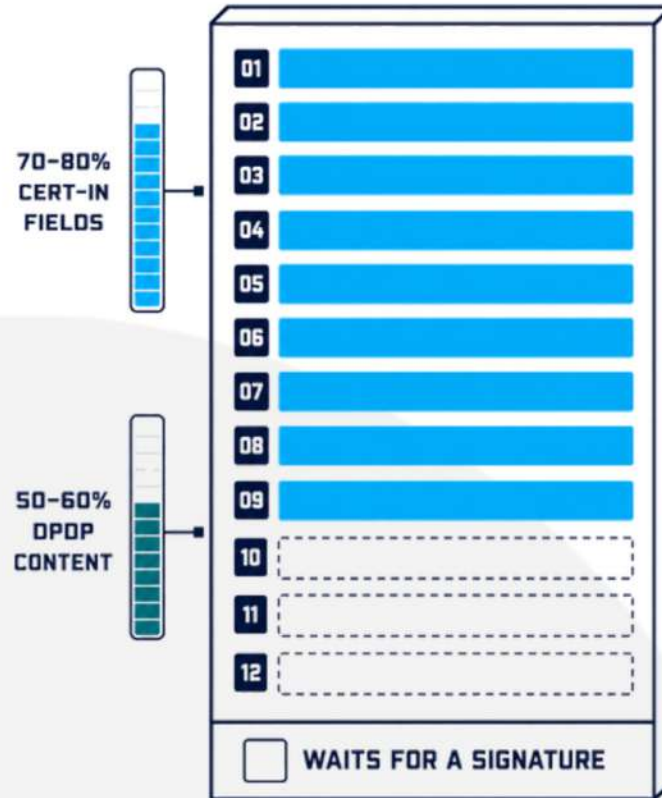


13 MAY 2027

RS 250 CR PLUS RS 200 CR



70-80% OF FIELDS PRE-FILLED



DIGITAL INDIA
UMBRELLA

CYBER SURAKSHIT BHARAT
1,637 CISOS TRAINED

C3IHUB IIT KANPUR
RS 30 LAKH COHORT

I4C
NATIONAL COORDINATION



31 JULY 2026
24X7 MANDATED
EFFECTIVE IMMEDIATELY

WHO IT AFFECTS

The IT head of a cooperative bank or small NBFC - usually the entire security team. RBI requires a **24x7 cyber centre** with six-hour reporting. Same shape at SEBI-regulated small firms and hospitals.

HOW WE FILL THE GAP, PRACTICALLY

The AI absorbs the routine alert work one person cannot carry. Every action is checked against a rulebook the IT head can read and change. On a confirmed incident, the record drafts the **regulator's report for a human to sign**. Sold through managed providers, so the bank builds nothing.

EXISTING SCHEMES

Digital India · Cyber Surakshit Bharat (1,837 CISOs trained - our buyers) · C3iHub IIT Kanpur (₹30 lakh, no equity) · I4C, MHA.

SCALE OF IMPACT



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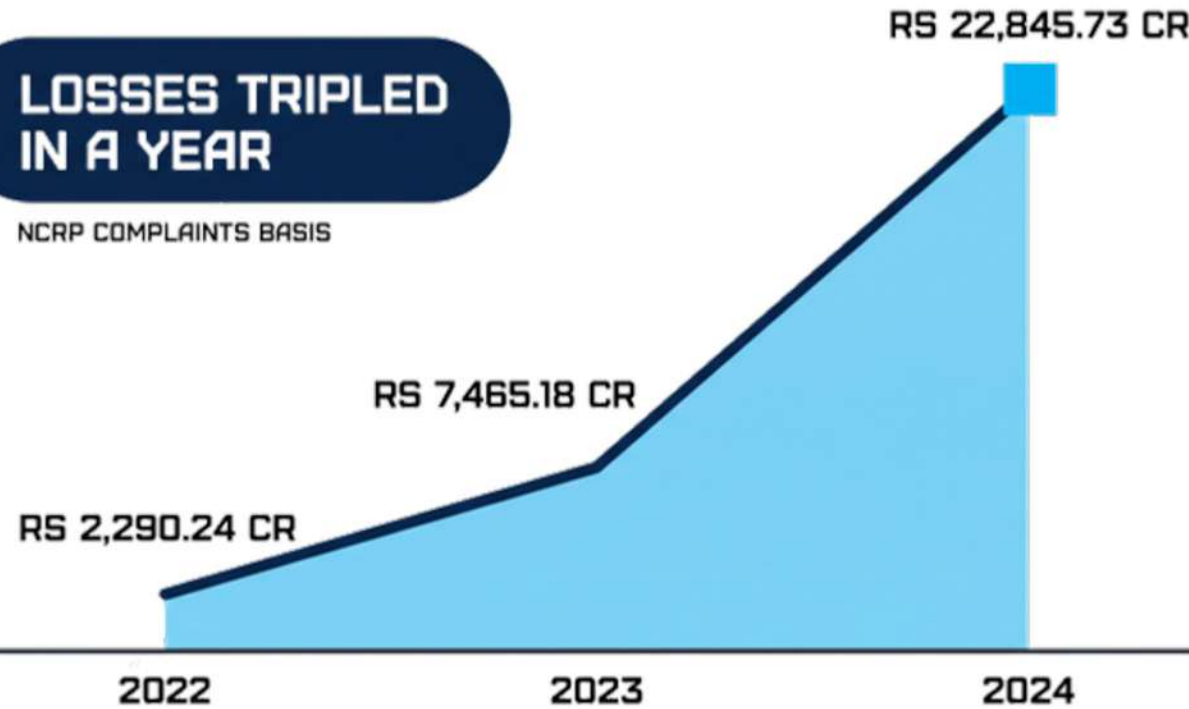


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six hours to report. ₹250 crore of exposure landing by 2027. 790,000 defenders India does not have.

LOSSES TRIPLED IN A YEAR

NCRP COMPLAINTS BASIS



TARGET USER SCORE

CO-OP BANKS AND NBFCs

19/20

HOSPITALS

17/20

SEBI SMALL ENTITIES

17/20

73%

LIMITED OR NO AI AUTOMATION

USD 2.2M

SAVED PER BREACH, GLOBAL

ALERTS ARRIVE

15 TO 40 MIN EACH
TRIAGE QUEUE

ACTUALLY WORKED

20% TO 40% NEVER SEEN

61% FASTER ON IDENTITY



LOSSES TRIPLED IN A YEAR

₹2,290 cr → ₹7,465 cr → ₹22,845 cr, 2022–2024.

Counted as citizen complaints on NCRP.

19/20 ON PAIN AND URGENCY

73% UNDER-USE AUTOMATION

40% NEVER INVESTIGATED

SELL ONCE, PROTECT MANY



WHERE ALERTS GO TO DIE

15–40 minutes each. A fifth to two-fifths are never seen

SELL ONCE, PROTECT MANY

One provider integration reaches hundreds of banks.

73% OF INDIAN ORGANISATIONS

BARELY USE AI IN SECURITY

The tool exists. Almost nobody has switched it on.

PROPOSED SOLUTION



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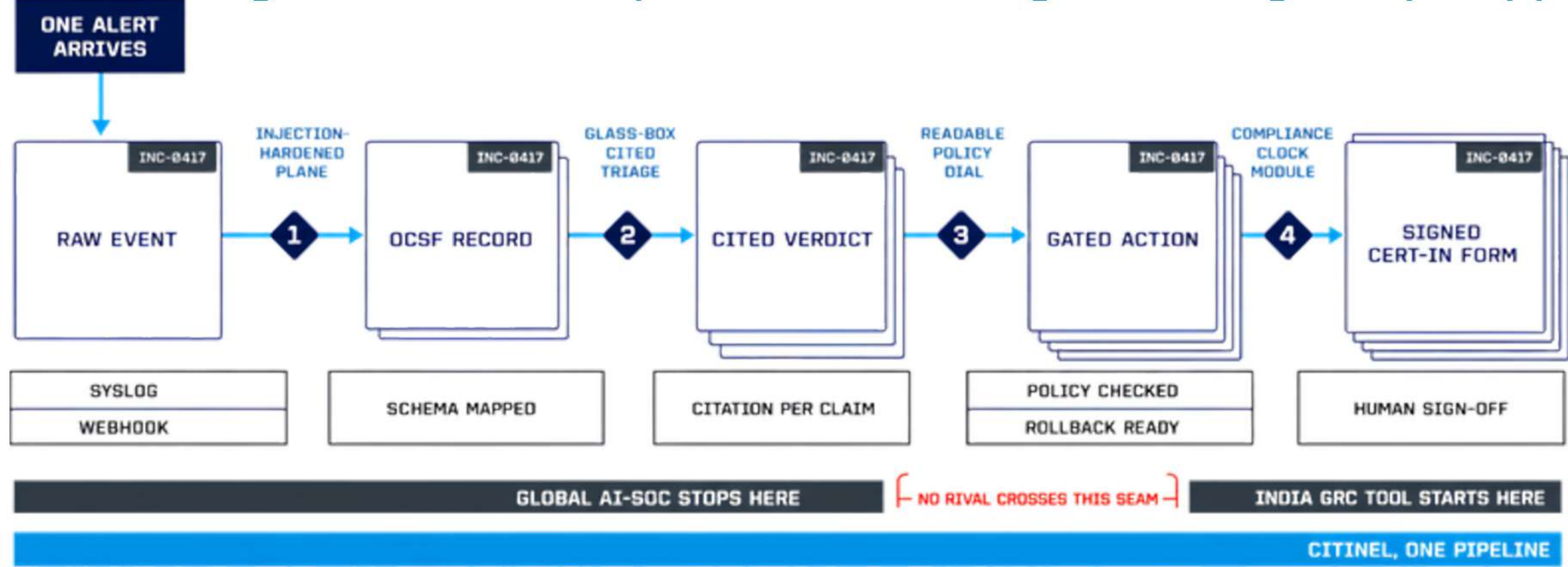


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the investigation writes the report. detection to regulator-ready draft, one pipeline.



CITINEL is an autonomous Cyber SOC that investigates every alert with cited evidence, acts only inside a policy humans can read, and beats India's regulatory clocks. Alerts from bank systems normalize to OCSF; Sigma rules catch known threats deterministically; Claude agents correlate the rest into a MITRE ATT&CK kill-chain narrative where every verdict cites the exact log line proving it. Response actions pass a per-action policy: enrichment runs automatically, host isolation runs with rollback, disabling a production account waits for one-click human approval. The pipeline treats all log content as untrusted data, so a poisoned log cannot hijack the AI. On confirmation, it drafts the CERT-In 6-hour report and DPDP breach artifacts for human sign-off. Built for cooperative banks that RBI requires to run a 24×7 SOC they cannot afford.

1 · THE COMPLIANCE CLOCK

The regulator's form fills itself. A human signs it.

2 · POISON-PROOF BY DESIGN

Fake instructions hidden in logs stay data.

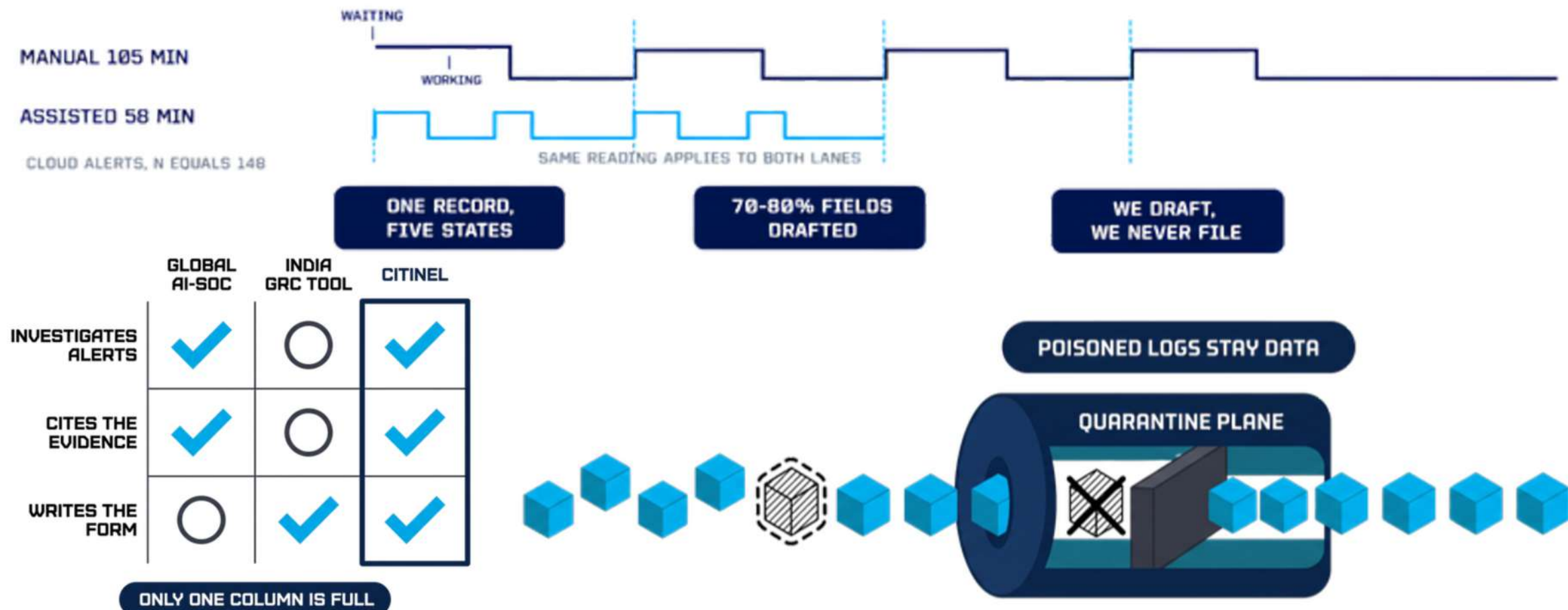
3 · A DIAL, NOT A SWITCH

Look up: auto. Isolate: auto with undo. Disable: waits.

4 · YOU KEEP WHAT IT LEARNS

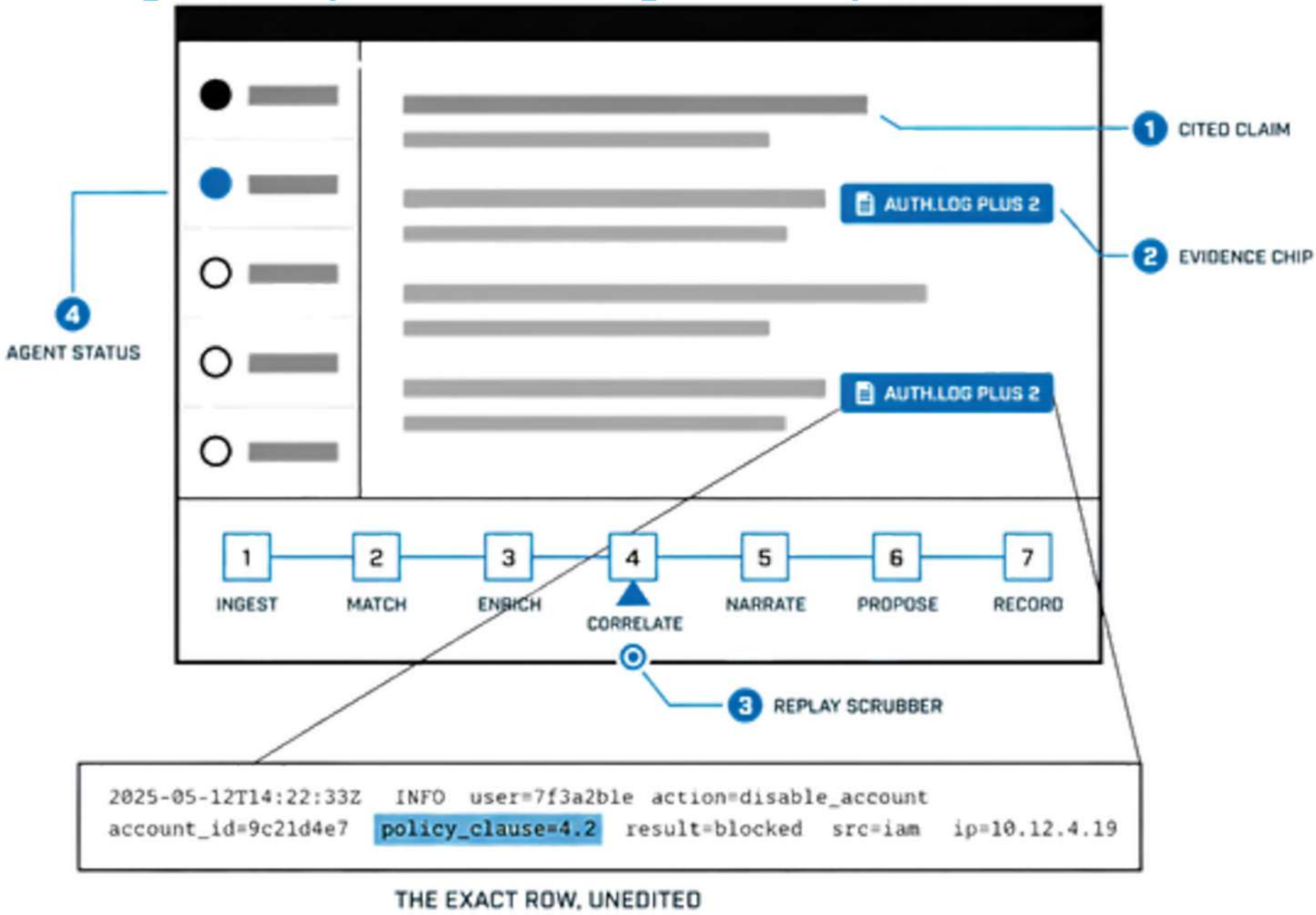
Every closed incident leaves you an open rule.

Global tools stop here. India's compliance tools start there.
We are the only one pipeline across.



USER EXPERIENCE

every verdict points at the log line that proves it. click it.



EVERY CLAIM CARRIES A CHIP 15 TO 40 MINUTES RECLAIMED

Every verdict points at the log line that proves it.
Click it.

NOTHING HERE ASKS THE ANALYST TO TRUST THE MODEL.

HANDOVER WRITES ITSELF

ACTION REQUIRES APPROVAL

DISABLE PRODUCTION ACCOUNT

ASSETS AFFECTED 14

POLICY CLAUSE 4.2

APPROVE DENY

APPROVE WITH BLAST RADIUS

ACTION REQUIRES APPROVAL

DISABLE PRODUCTION ACCOUNT

ASSETS AFFECTED 14

POLICY CLAUSE 4.2

APPROVE DENY

SHIFT HANDOVER

- OPEN INCIDENTS
- AWAITING APPROVAL
- ACTIONS TAKEN
- WATCH LIST

GENERATED AT SHIFT END

SUPPORTS AND ARGUES AGAINST

CONFIDENCE 75%

SUPPORTS ARGUES AGAINST

CONFIDENCE, NOT CERTAINTY

WE DELETED A GOOD LINE

SHIFT HANDOVER

OPEN INCIDENTS

AWAITING APPROVAL

ACTIONS TAKEN

WATCH LIST

GENERATED AT SHIFT END

IT SHOWS THE DOUBT

CHIP OPENS THE PROOF.

RETIRED

CHECKED, THEN CUT

KEPT

ANALYST VIEW

CISO VIEW

SAME RECORD

RAW RECORD, UNEDITED

TECH STACK & ARCHITECTURE



rules before models. policy before actions. citations before conclusions.

RUNTIME

TAVILY
LIVE OSINT FOR ENRICHMENT
CITED IOC AND ACTOR CONTEXT

SWYTCHCODE
AGENT TO API EXECUTION
TICKETING PLUS COMMS, TWO APIS

N8N
EXPORT AND ESCALATION FLOWS
OPA REMAINS THE AUTHORITY

RENDER
WEB SERVICE PLUS WORKER
POSTGRES AND REDIS, THREE TYPES



BEST-USE CRITERIA MET

BUILD

CODEMATE
AI PAIR BUILD, NOT RUNTIME



GO-TO-MARKET

STARTUPED
CHANNEL-FIRST B2B2X NARRATIVE



Rules before models. Policy before actions. Citations before conclusions.

- TAVILY** : Live threat intel from the open web, pulled in as citable evidence.
- SWYTCHCODE** : The path from agent to real API. Response and report agents act through it.
- n8n** : Escalation, notification and export flows. OPA stays the authority.
- RENDER** : Three service types: web app, background worker, database.
- CODEMATE.AI** : AI development agent used while building.
- STARTUPED** : Validation, market research and GTM. Slide 13 is built from it.

SIX PARTNERS, ZERO DECORATION

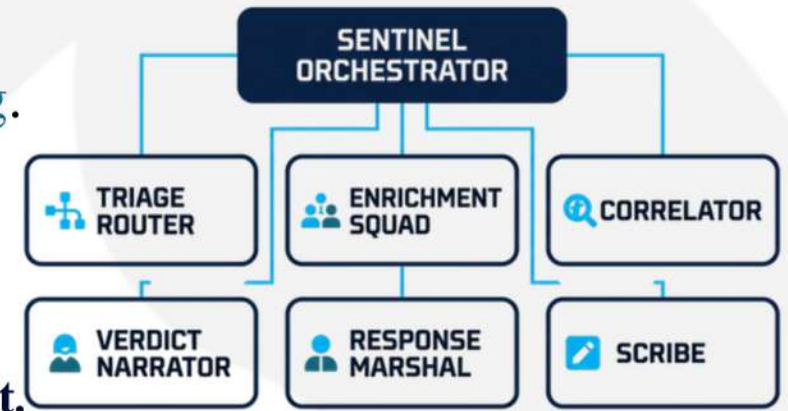
Remove any one and a stage of the pipeline stops working.

3,000 KNOWN-THREAT CHECKS RUN FIRST

Cheap, instant, same answer every time. AI is asked last.

15× THE COST, SPENT ON PURPOSE

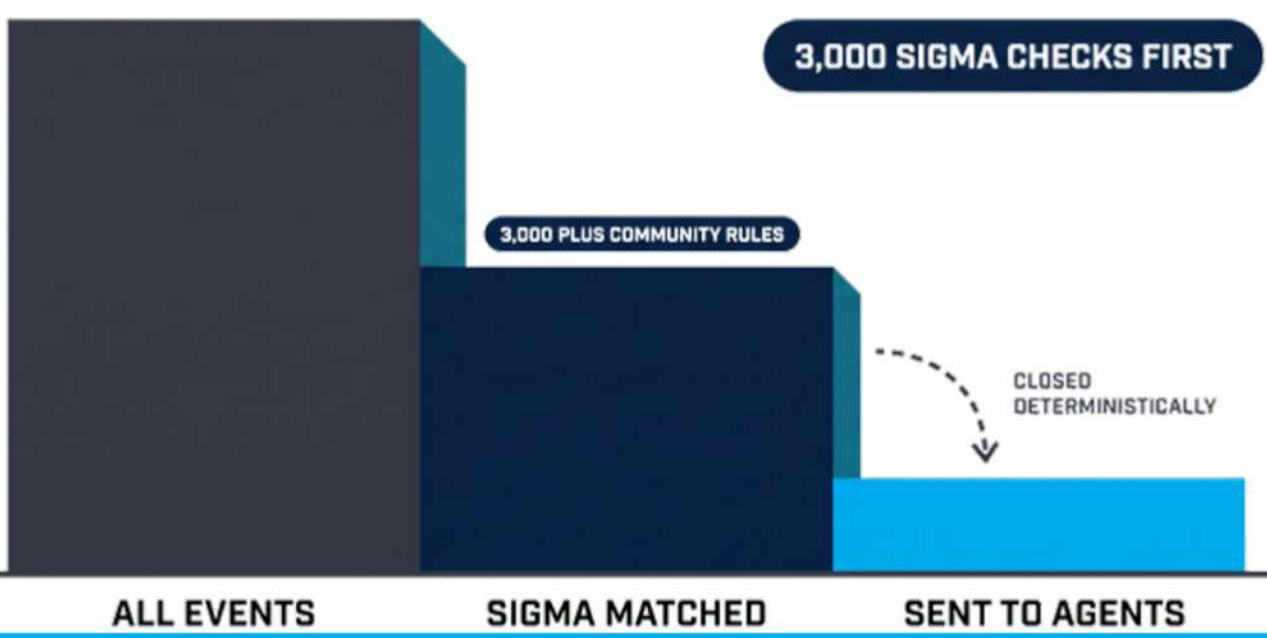
An agent costs ~15× a rule check. So we ask it only when it earns it.



SEVEN AGENTS, ONE RECORD

SPAWNED PER INCIDENT

SIX PARTNERS, ZERO DECORATION



EVERY LAYER EARNS ITS PLACE

SEVEN AGENTS, ONE JOB EACH

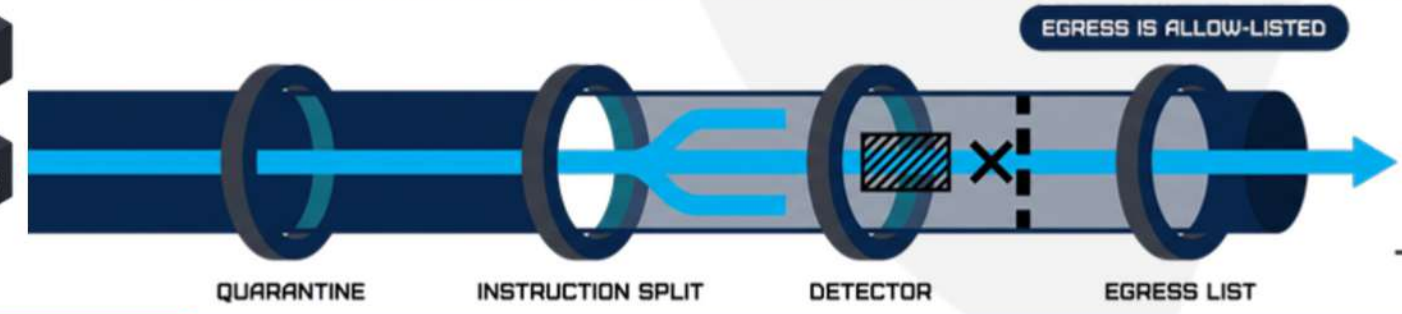
Sentinel splits the incident · Router sorts by severity · Enrichment looks things up · Correlator builds the attack story · Narrator writes it with citations · Marshal proposes the action · Scribe drafts the report.

15X COST, SPENT ON PURPOSE

15X TOKENS

WHY THE NARRATOR IS SEPARATE

Citations and rigid output formats can't come from the same call.



1X TOKENS

SINGLE WORKFLOW

AGENT SWARM

ONLY WHERE BREADTH PAYS

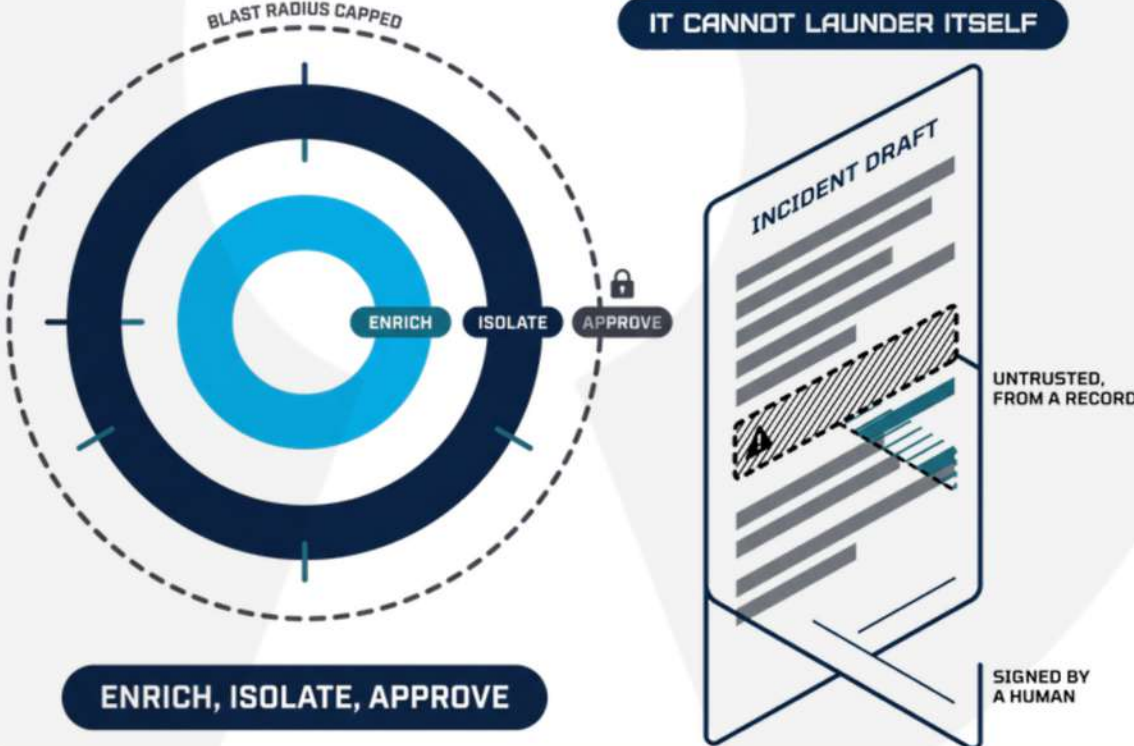
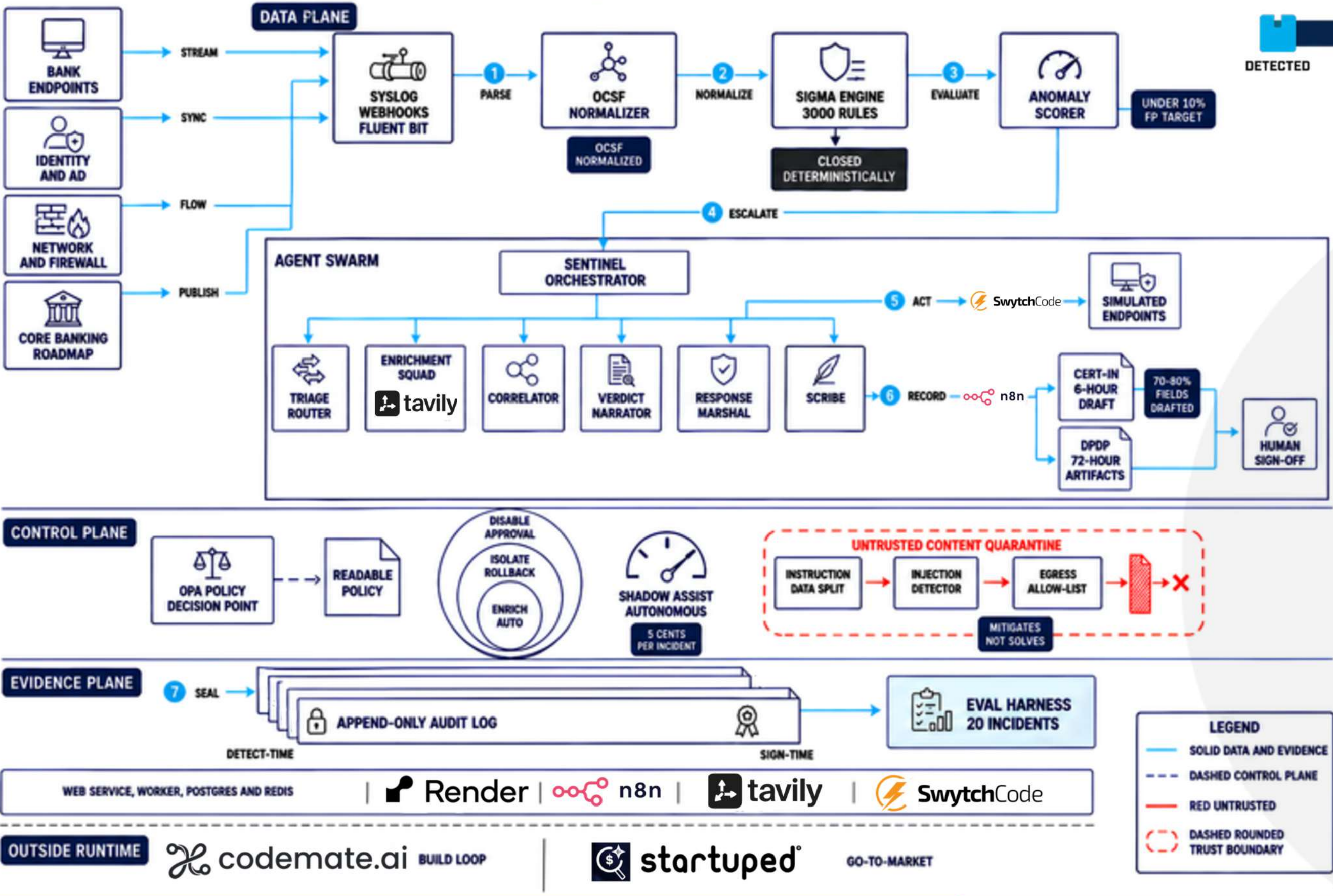
one dial decides what runs automatically, what runs with rollback, and what waits for a human.

APPEND-ONLY, NEVER EDITED



THE QUARANTINE PLANE

Log content enters as data and never leaves as an instruction.
The red X is an attacker's hidden command being stopped.



FEASIBILITY



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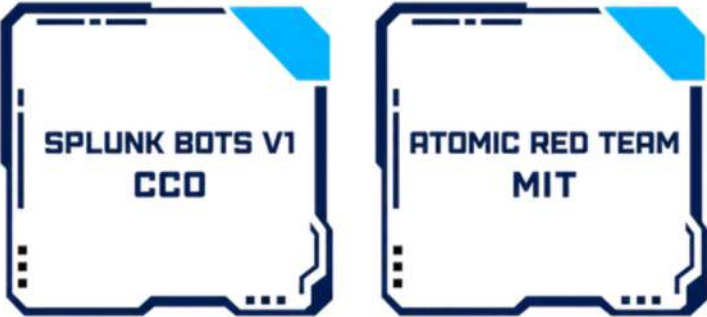


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three stages, one published out-of-scope list, and a pre-committed fallback for each.



... ISOLATED VM ONLY ...

CCO AND MIT, LICENCE-CLEAN

WHAT WE ARE NOT BUILDING.

- ☐ MULTI-TENANCY
- ☐ HONEYPOTS
- ☐ MODEL FINE-TUNING
- ☐ MACHINE FILING
- ☐ FULL ATT AND CK COVERAGE

ROADMAP, NOT PRETENCE.

De-Risking

THE RISKY PARTS ARE ALREADY PROVEN

Log format, detection rules, policy engine and regulator form fields have all been exercised. What is left is assembly.

WE BUILD THE HARD PART OURSELVES

The novel piece the regulator's draft written straight from the incident record is the piece we own end to end.

Data Provenance

LICENCE-CLEAN DATA, ISOLATED MACHINE

Public security datasets under CC0 and MIT licences, replayed inside an isolated VM. No customer data. No legal exposure.

The Three Stages

THREE STAGES, EACH DEMOABLE ALONE

Stage 1 : logs in, rules running.

Stage 2 : agents, policy gate, audit trail.

Stage 3 : report drafter, glass-box screens, attack demo, scoring.

The Measurement

OUR TARGET: UNDER 10% FALSE POSITIVES

Measured industry range on labelled data: 40% to 80%. It stays a target until our harness publishes the number.

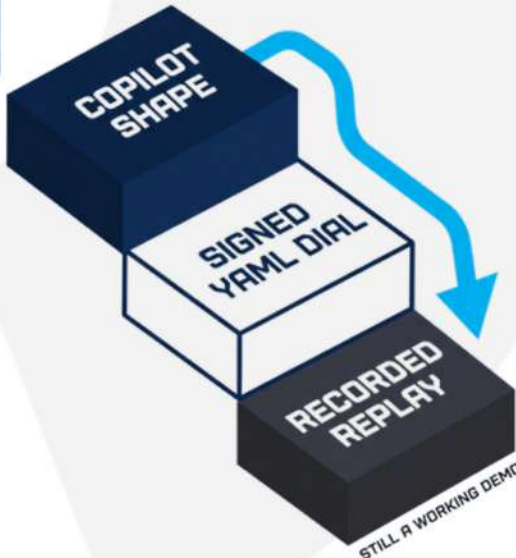
MEASURED ON 20 LABELLED INCIDENTS.

THE RISKY PARTS ARE PROVEN



WE BUILD THE NOVEL PART
THE CERT-IN DRAFTER

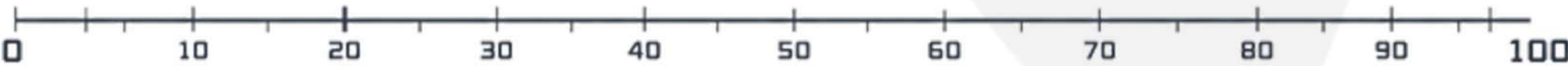
IF IT SLIPS, IT DEGRADES



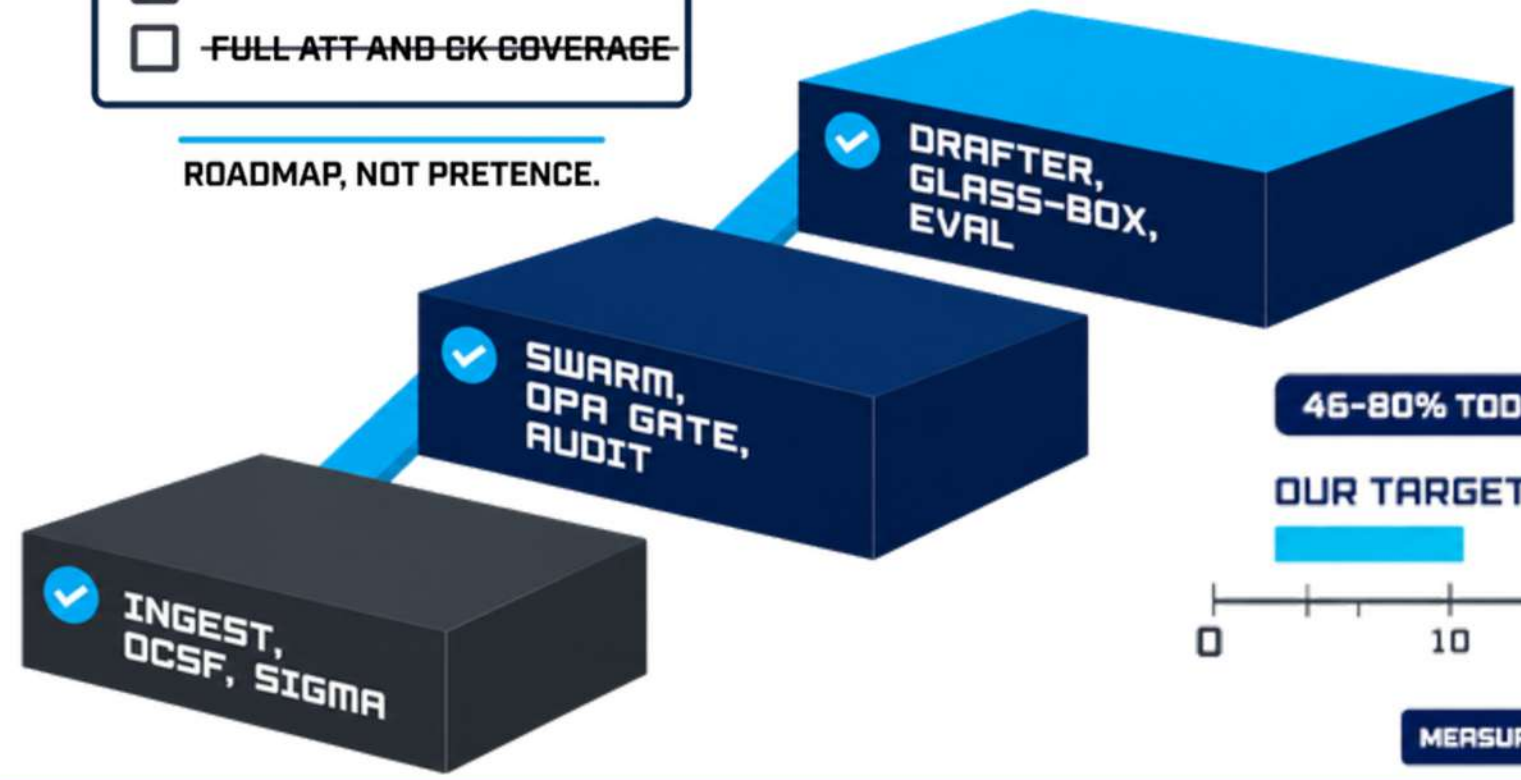
46-80% TODAY, UNDER 10% TARGET

OUR TARGET

INDUSTRY RANGE



MEASURED ON 20 LABELLED INCIDENTS



STILL A WORKING DEMO

PRACTICABILITY



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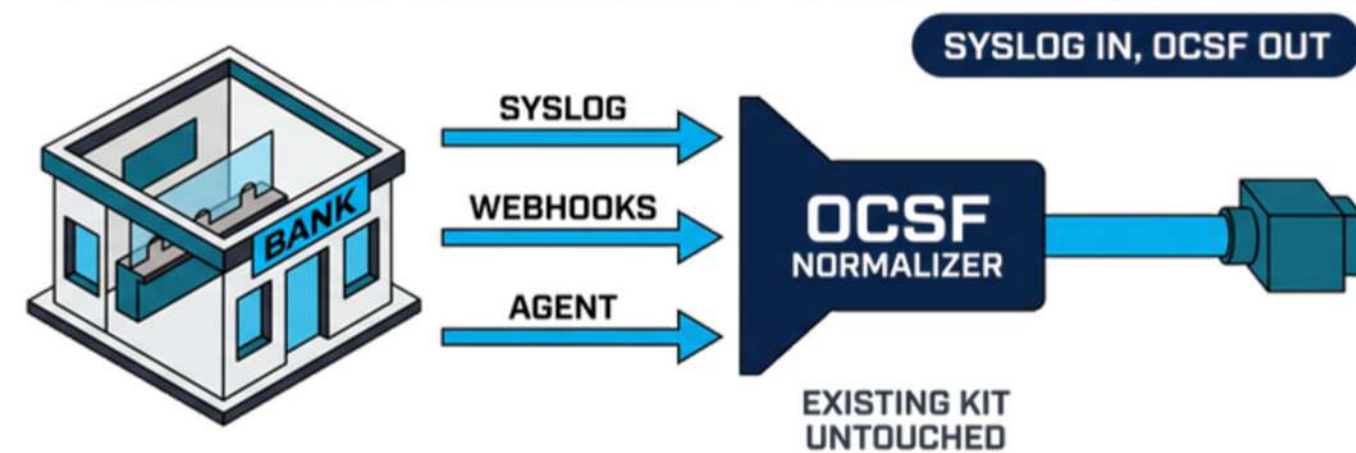


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rules handle the known threats cheaply. when feeds fail, it degrades to deterministic detection instead of guessing.



Rules handle the known threats cheaply. When feeds fail, it degrades to deterministic detection instead of guessing.

IT SITS BESIDE WHAT THE BANK ALREADY RUNS. NOTHING GETS RIPPED OUT.

THREE INPUTS, ONE LANGUAGE

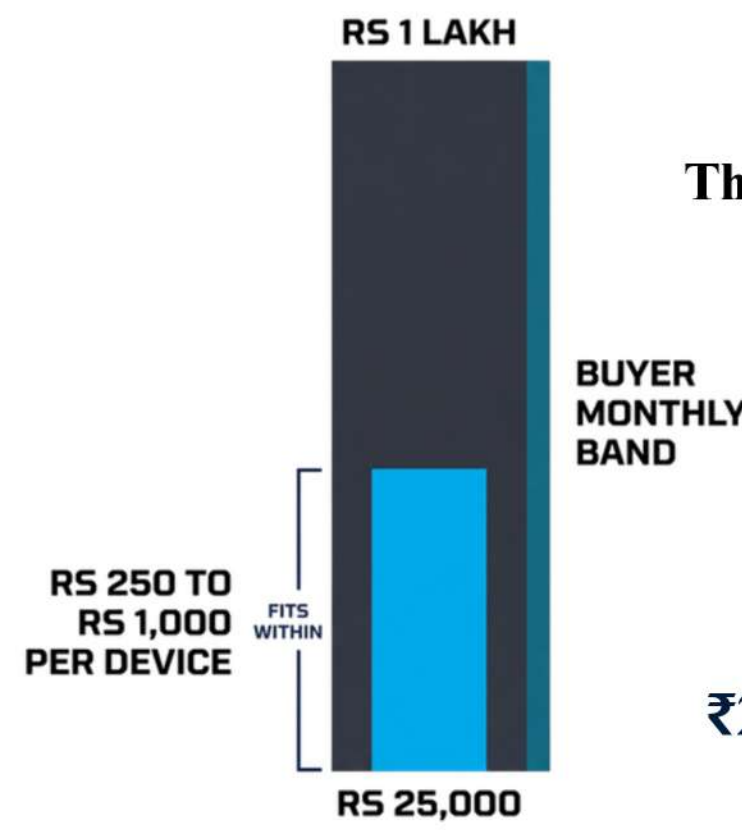
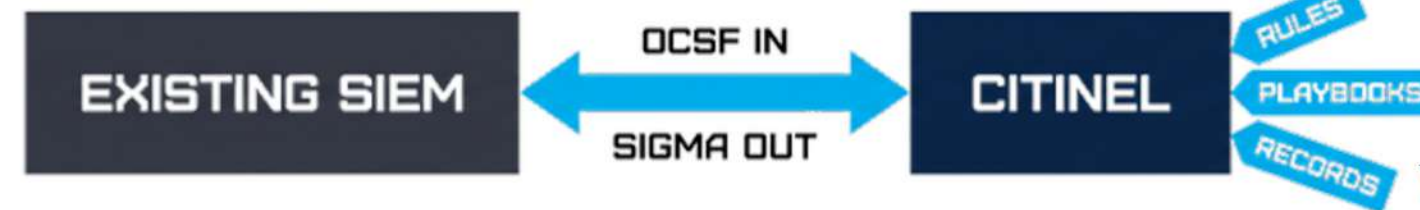
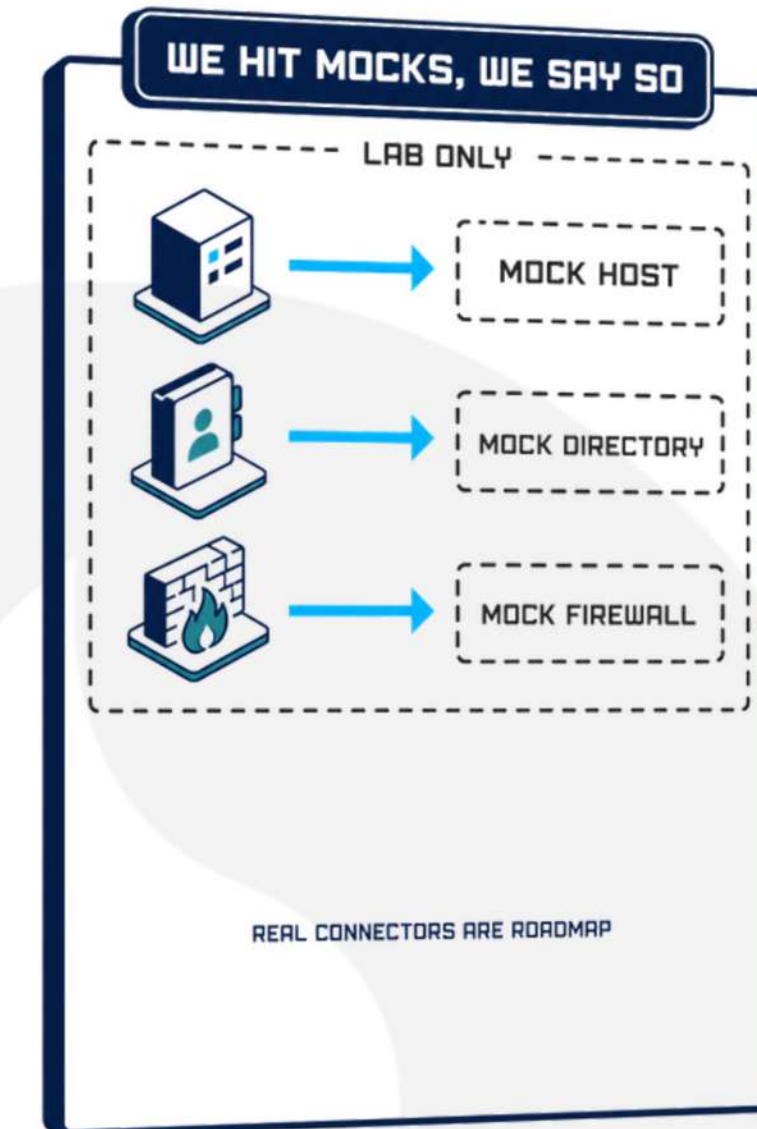
Syslog, webhooks, agent. A new source is a mapping, not a rebuild.

EXISTING KIT UNTOUCHED

No software on every machine. No system replaced. No downtime.

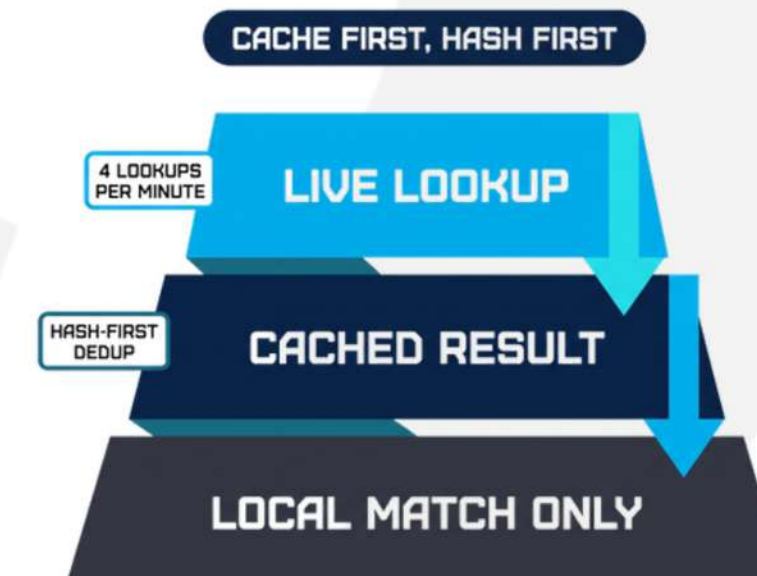
IT SITS BESIDE YOUR SIEM

We read what your system already sends out, and hand back detection rules it can run.



THE 2 AM TEST
The hour when nobody is awake. Every rung shows what runs without a human, and exactly where it stops and waits.
SOLID BAR RUNS. HOLLOW BAR WAITS.
24 MINUTES INSIDE THE CLOCK
Six-hour mandate. Twenty-four minutes used.

INSIDE THE EXISTING BILL
₹250 to ₹1,000 per device, inside a ₹25,000 to ₹1 lakh monthly budget. It fits the bill the bank already pays.



34 MINUTES INSIDE THE CLOCK

INSIDE THE EXISTING BILL

SUSTAINABILITY

open standards, portable rules, and a channel the regulator already built for us.



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OTHERS PATCH OUR DETECTIONS.

BESPOKE UPKEEP

3,000 PLUS SHARED CHECKS

COMMUNITY UPKEEP

COVERAGE OVER TIME

WHEN A GIANT COPIES US



PRICE PHYSICS
ENTRY USD 36,000
BUYER RS 25K A MONTH



MANDATED CHANNEL
SEBI BUILT THE DOOR
WE WALK THROUGH IT



OPEN-ARTIFACT TRAP
COPY OUR EXPORTS
LOCK-IN DIES

STRUCTURAL, NOT SECRET

Open standards, portable rules, and a channel the regulator already built for us.

TAKE YOUR DETECTIONS AND GO

Every incident we close produces an open rule in a standard format.

If you leave, you keep them. We make leaving easy on purpose that is why people stay.

OTHERS CAN RUN OUR DETECTIONS

Structural, not secret. The moat is the channel, not the code.

EVERY CLOSE FEEDS THE NEXT

Incident closed → rule drafted → human reviews → library grows → the next incident is caught faster.

TAKE YOUR DETECTIONS AND GO



SIGMA
RULE

PLAYBOOK

AUDIT
RECORD

OPEN FORMATS, NO EXIT FEE

CORPUS
GROWS

PORTABLE
SIGMA

RULE
DRAFTED

HUMAN
REVIEW

INCIDENT
CLOSED

EVERY CLOSE FEEDS THE NEXT

RS 30 LAKH, NON-DILUTIVE

C3IHUB IIT KANPUR

RS 20 LAKH GRANT PLUS RS 10 LAKH

49 STARTUPS NURTURED

60% PLUS COMMERCIALISED

CYBER SURAKSHIT BHARAT

1,637 CISOS ACROSS 44 BATCHES

NO EQUITY GIVEN UP

WAZUH

30M PLUS
DOWNLOADS A YEAR

UNFUNDED
SINCE 2015

PAID CLOUD
AND SUPPORT

30M DOWNLOADS, ZERO FUNDING

THEHIVE TO STRANGEBEE

OPEN SOURCE
FROM 2014

COMMERCIAL
FROM 2022

EXPLICIT
MSSP TIER

NATIONAL ALIGNMENT, SDG TARGETS

8.10

DOMESTIC BANKS
STRENGTHENED

9.1

RESILIENT
INFRASTRUCTURE

16.6

ACCOUNTABLE
INSTITUTIONS

1.5

SHOCK-PROOF
SAVINGS

FUTURE PROGRESSION: BUSINESS MODEL & ROADMAP



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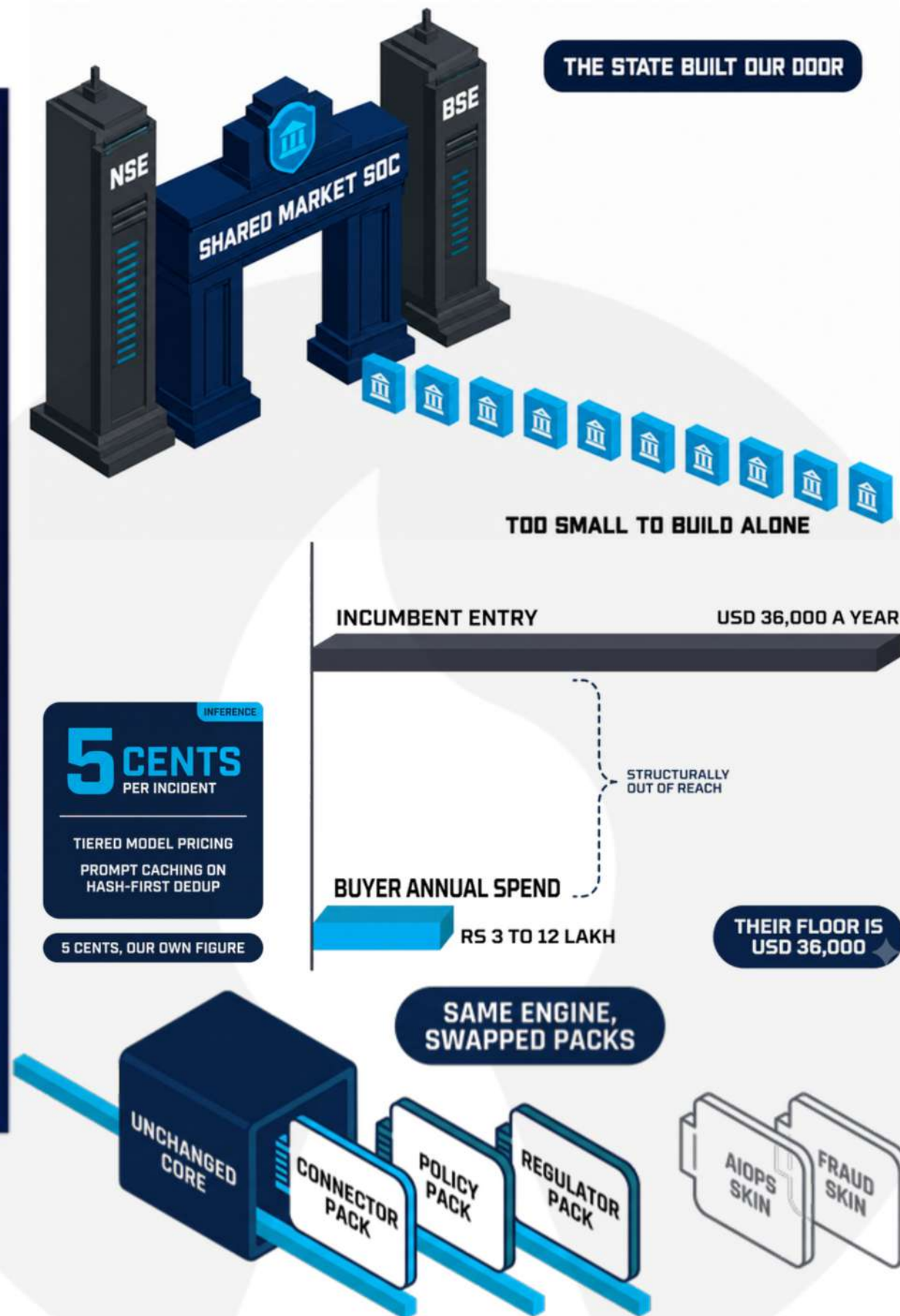
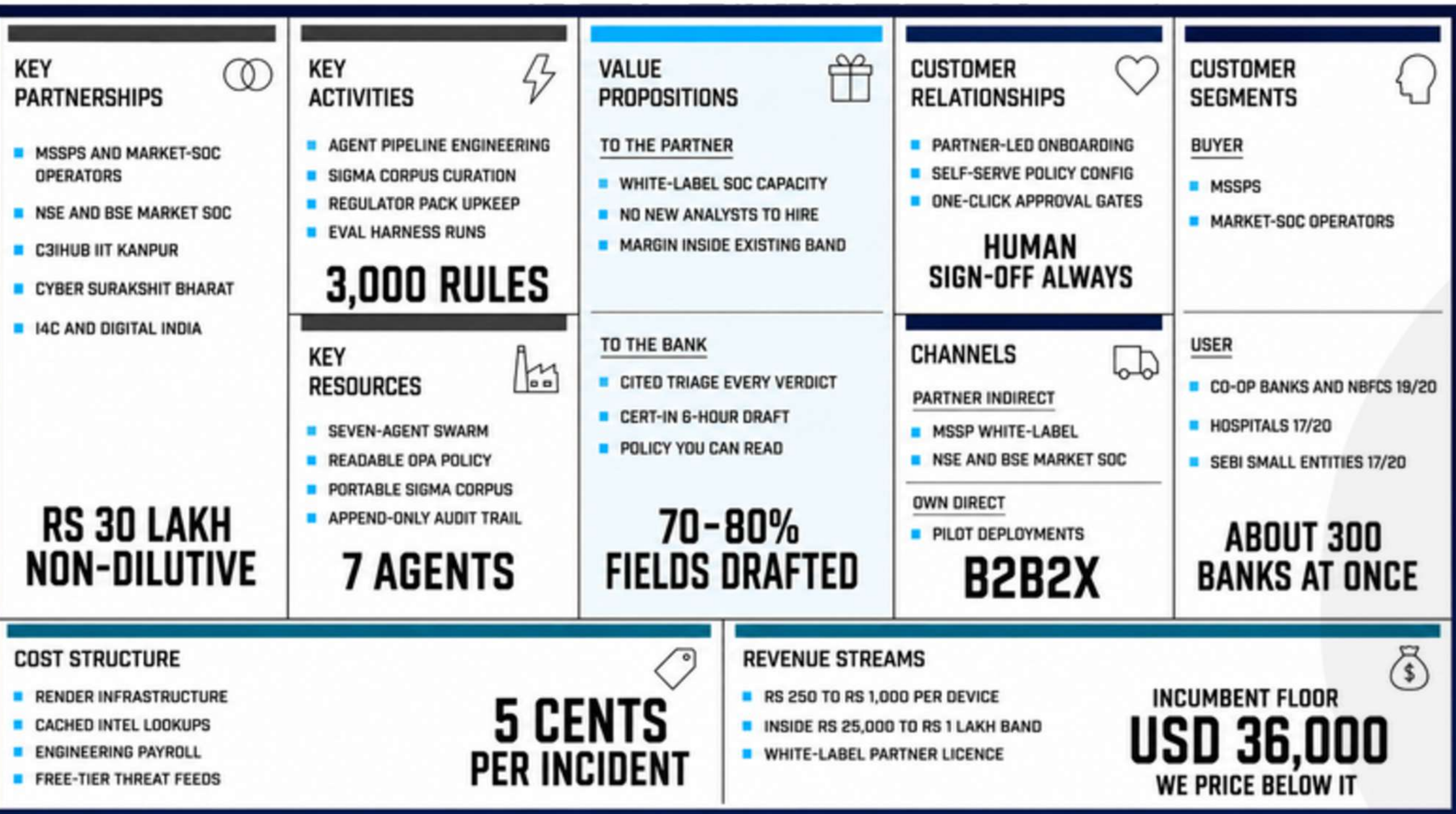


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sold through mssps, not to banks. a new regulator is new config, not new code.



VALUE PROPOSITION

Proven verdict every time. No analysts to hire.

Report drafted before the clock runs out.

Sold through MSSPs, not to banks.

A new regulator is new config, not new code.

THANK YOU

scan the code. the prototype is already running.



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EVIDENCE



POLICY



CLOCK

EVIDENCE, POLICY, CLOCK

THREE MINUTE WALKTHROUGH



WATCH NOW

SKIP

THREE MINUTE
WALKTHROUGH
LIVE

THREE MINUTE
WALKTHROUGH
BRIEF

THREE MINUTE
WALKTHROUGH
VIDEO



THE WATCH CONTINUES



AUTONOMY IS A DIAL

ASK US THE HARD ONE

- ✓ WHEN IT IS WRONG
- ✓ HALLUCINATION
- ✓ FALSE POSITIVE RATE
- ✓ WHO ACTUALLY PAYS

WE HAVE ANSWERS
FOR ALL FOUR



EVIDENCE & REFERENCES

every number above, with its source, its date, and its denominator.



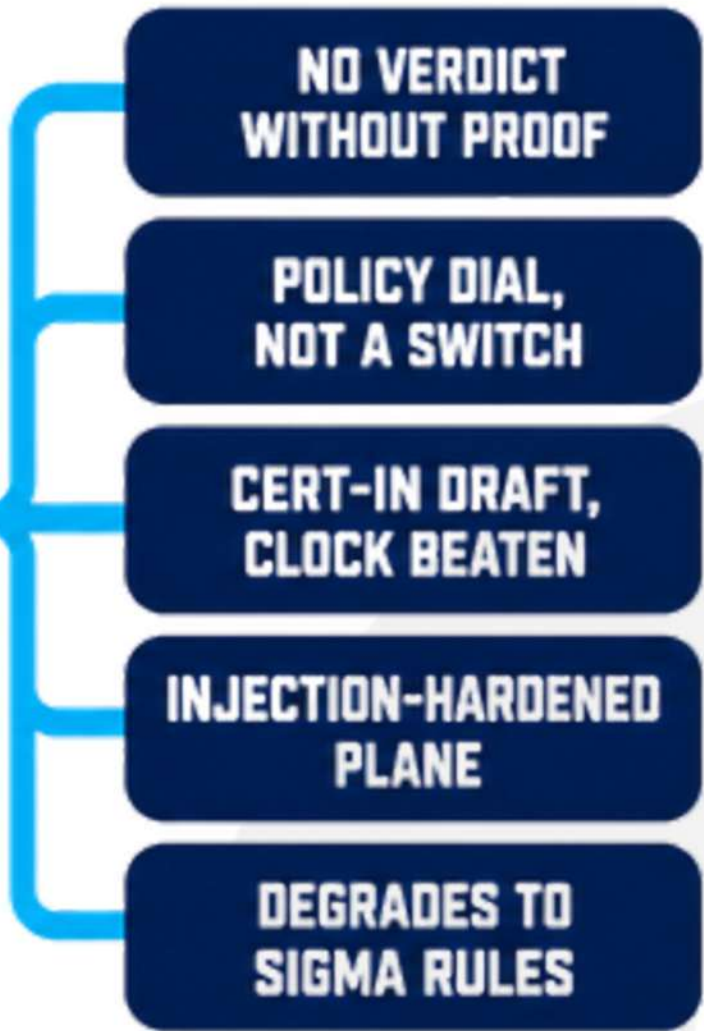
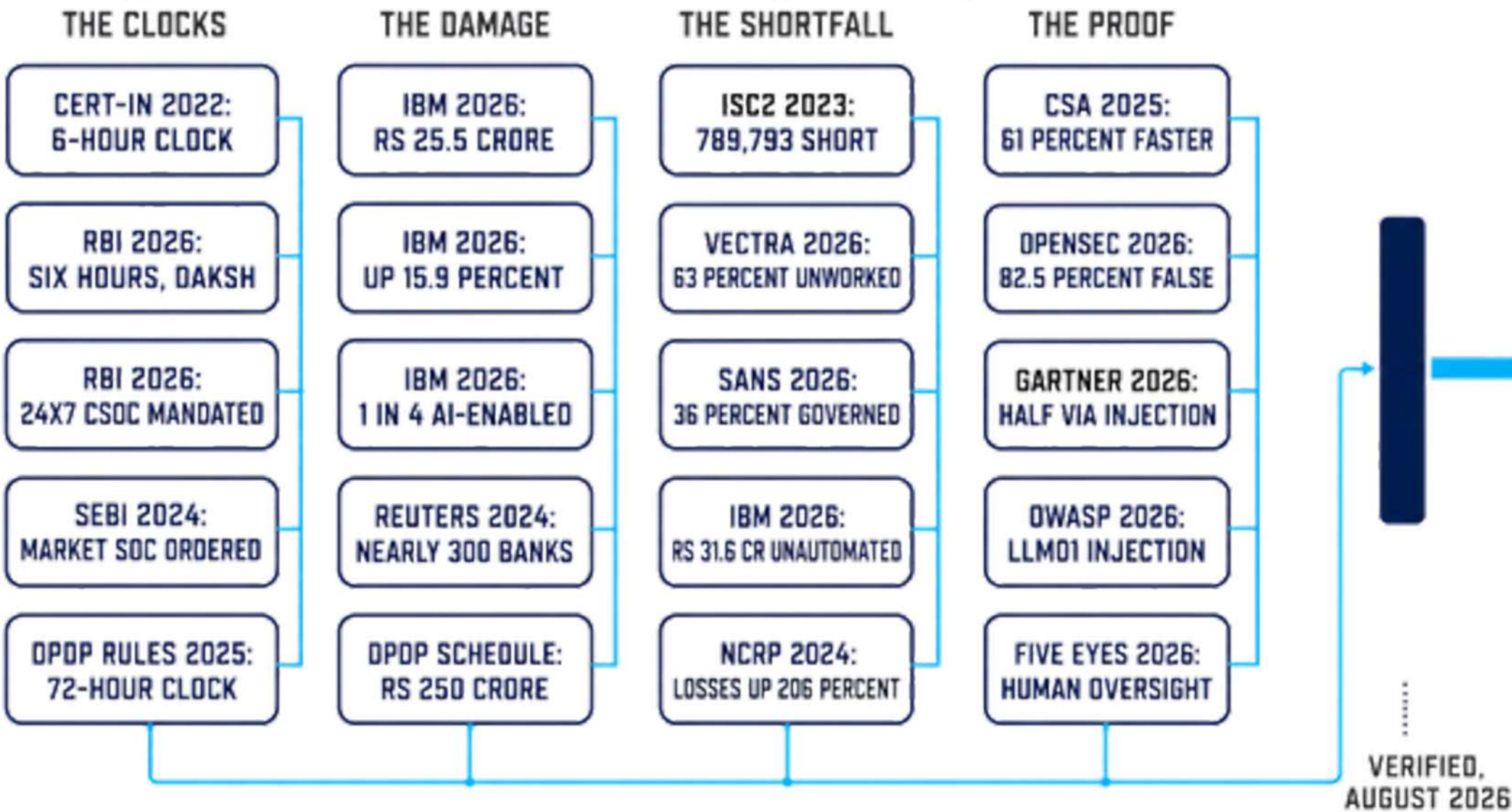
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HIGH

MED

LOW

INFERENCE

TWO IBM STUDIES, NOT ONE

RATES ARE RANGES

TWO COUNTS, NEVER MERGED

DPDP IS A RUNWAY, NOT A LIVE PENALTY

THE ATTACK IS PEER-PUBLISHED



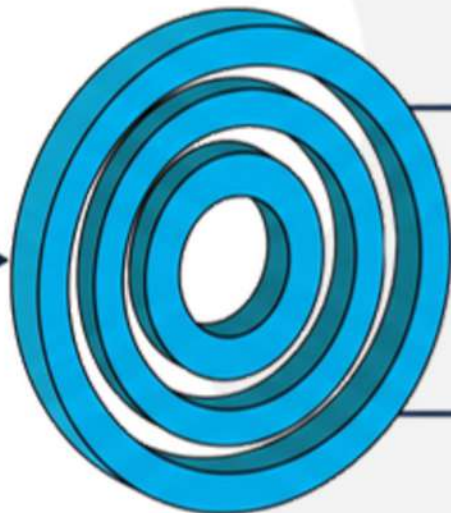
PROMPT INJECTION IN SYSTEM LOGS
ARXIV 2607.24174

WE DEMO IT LIVE



OCSF NORMALIZER

SIGMA GATE



SIMULATED ENDPOINTS

CERT-IN DRAFT CARD

APPEND-ONLY AUDIT